

ALPACA AI TRADING AGENTS HACKATHON · LABLAB.AI × ALPACA

Theta Council

An autonomous options desk on Alpaca.
Five agents, one veto, a full audit trail.

paper PA3YGPXXKC9E

start \$100,000

options level 3

0 dependencies

52 tests

Most “AI trading agents” hand a model an order endpoint and a prompt that says *be careful*.

Models are **excellent** at reading context across a dozen instruments at once.

Models are **unreliable** about position size on the fortieth API call of the session.

So don't ask them to be.

The LLM has a veto and a dial. It never has the wheel.

IT CAN

reject any trade, with a reason

lower conviction → lower size

set a posture that only tightens risk

IT CANNOT

invent a strike

raise a quantity

widen a risk limit

reach an order endpoint

`size_multiplier` is clamped to ≤ 1.0 . There is no way to ask for more size.

Selling an option is selling a *forecast of movement*.

The market prices that forecast at **implied volatility**. The underlying then delivers **realised volatility**. On average the first is larger — a fee paid to whoever carries the gamma risk, not a prediction about direction.

```
vrp_ratio = ATM implied ÷ 20-day realised
```

```
≥ 1.12    the market overpays for movement    → SELL premium
```

```
≤ 0.98    movement is cheap                    → BUY  premium
```

```
between   → NO TRADE
```

The third branch is where a real desk spends most of its life — and the branch nobody demos.

In dollars, not adjectives.

V_{iv} at market implied vol → what we get paid

V_{rv} at 20-day realised vol → what we think it is worth

spread we **SELL**: EV per share = credit - V_{rv}

spread we **BUY** : EV per share = V_{rv} - debit

edge_ratio = $EV \div \max_loss$ ← the ranking statistic

Both legs, same measure, same horizon — so the difference **is** the variance premium in dollars.

The credit is already haircut for the bid/ask we expect to cross, so the edge is quoted **after** friction.

EV not positive → the structure is never built.

THE COUNCIL

AGENT	JOB	CAN IT LOSE MONEY?
Analyst	Regime in ATR units, so SPY and NVDA are comparable. Flags vol shocks.	<i>no</i>
Vol Scout	The sell / buy / stand-aside verdict, plus a 0–1 confidence.	<i>no</i>
Architect	Real strikes from the live chain; prices its own EV after friction.	<i>no</i>
Adjudicator (LLM)	Cross-sectional judgement. Veto and size dial only.	<i>only downward</i>
Risk Officer	24 deterministic gates, sizing, kill switches. Holds the veto.	it is what prevents it
Position Manager	Exits. Runs before entries, every cycle.	this is where the P&L is
Bandit	Beta posterior per regime × structure → feeds back into size.	<i>no</i>

Arithmetic can't see these three things.

Three of today's candidates are **the same index bet in three costumes**.

A 1.4× premium across **every megacap at once** is a macro print being priced, not five independent gifts.

The book has quietly become **one-way short puts**.

Guardrails. Invented candidate ids are dropped and logged. **Silence is not consent** — a candidate the model doesn't mention is discarded. On failure or unparseable output the desk continues deterministically and journals that it did.

ONE CYCLE

01 Intake

One brief, shared by every agent, journalled verbatim.

03 Manage — before entries

Closing a winner returns the risk budget a new entry then uses.

05 Design

Real strikes from the live chain; EV priced after friction.

07 Review

Risk Officer gates each survivor and sets the size.

02 Read

Analyst forms the regime view; Vol Scout the verdict.

04 Portfolio gates

Kill switches, delta band, buying power, session clock.

06 Adjudicate

The model may veto and may shrink. Nothing else.

08 Execute

Single **mleg** orders — then journal everything, refusals included.

A desk that opens before it closes drifts to its position cap and sits there.

PASS market_open — market open

PASS account_tradable — ACTIVE

PASS options_level — level 3

PASS daily_loss_kill — +0.00% vs -2.50%

PASS drawdown_kill — +0.00% vs -8.00%

PASS bp_reserve — 100%, floor 25%

PASS position_count — 0 of 6

PASS net_delta_band — +0 vs ±150

PASS deployed_risk — 0.0%, cap 12%

PASS time_of_day — 180 min left

Fourteen more per candidate — and sizing is one of them, not an afterthought:

positive_expectancy

credit_to_width

short_delta_band

prob_touch

leg_liquidity

per_underlying_cap

earnings_blackout

sizing

```
qty = floor( min(per_trade_budget, book_headroom) × llm_conviction ÷ max_loss )
```

TIME The last quarter of the trade's own life, capped at 7 days. A 32-day spread exits at 7 DTE; a 5-day spread at 1. A fixed rule would close a short-dated trade the moment it opened.

TARGET 55% of max profit. The last 45% takes most of the remaining time to earn and carries all of the remaining gamma.

STOP 2× the credit received, before max loss.

DEFEND Short-strike delta through 0.40 — this is no longer the trade that was approved.

Refusals are decisions too. Blocked candidates are journalled with the gate that stopped each one and its numbers. A desk that only logs its fills is grading its own homework.

Alpaca CLI

PREFERRED

JSON on stdout. Subcommands *probed* with `--help`, not assumed. The exact argv is journalled — replay any decision by hand.

Alpaca MCP

72 TOOLS DISCOVERED

stdio JSON-RPC. Reads `tools/list` and every tool's JSON schema, then binds intents to what the server actually offers. Late binding, not baked.

REST

GUARANTEED FLOOR

stdlib `urllib`. So the desk never stalls because a binary is missing on someone else's machine.

Spreads go over the CLI, not just REST. `alpaca order submit --order-class mleg --legs '[...]'` — and `--symbol / --side` must be omitted for mleg, because they're per-leg.

Credits are negative. The convention has shifted between releases, so the first credit order is sent with the documented sign and, on a price rejection, retried once flipped — then remembered and logged.

Every call records which pipe served it. “We used MCP and the CLI” is a table in the journal, not a claim in a README.

RUN IT YOURSELF

```
python -m council once --mock      # whole pipeline, offline, no keys, ~2s
python -m council mcp-doctor       # the discovered MCP tool map
python -m council preflight        # account, level, balance, feeds, pipes
python -m council loop             # trade the session
python -m council dashboard        # http://127.0.0.1:8787
python -m unittest discover -s tests # 52 tests
```

github.com/adeelsaleem844/Theta-Council

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MIT

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